Data-Driven Educational Quality Assessment: Student Risk Stratification via K-Means Clustering

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Abstract—**The unprecedented expansion of higher education in China has been accompanied by a paradox of high enrollment rates but questionable educational quality, where official statistics showing low dropout rates often mask deep-seated quality crises and "hidden dropout" phenomena. To address this, identifying at-risk students accurately is paramount. This study utilizes a dataset comprising 4,424 student observations to construct a dropout prediction model. We employ Cohen's d effect size for feature selection, identifying six critical attributes including academic performance and tuition status, and implement an optimized K-Means clustering algorithm where $K = 9$ is determined as the optimal number of clusters to reveal fine-grained student profiles. To validate the clustering quality, t-SNE is used to visualize the high-dimensional separation of the groups. The results successfully categorize students into High-Risk (Red) and Medium-Risk (Yellow) groups, proposing a targeted intervention framework that includes emergency financial and academic aid for high-risk students and curriculum planning support for medium-risk "stagnant" students. Consequently, this research provides a methodological foundation for an Early Warning System (EWS), shifting the focus from passive evaluation to proactive intervention in higher education management.**

Index Terms—**Educational Data Mining, Dropout Prediction, K-Means Clustering, Student Intervention, Higher Education Quality**

I. Introduction

The scale of higher education in China has achieved unprecedented expansion. By the end of 2023, the total number of students enrolled in higher education in China reached approximately 47.63 million, an increase of about 1.08 million compared to the previous year, with the gross enrollment rate surpassing 60% [1]. However, the "graduation rate myth" accompanying this high enrollment rate has sparked widespread controversy among the public. According to statistics from relevant scholars, the average graduation rate of ordinary undergraduate institutions in China (excluding "Double First-Class" universities) approaches 100%, which stands in sharp contrast to mainstream international quality assessment standards.

The *Education at a Glance 2022: OECD Indicators*, released by the Organisation for Economic Co-operation and Development (OECD) in 2022, reveals that OECD member countries such as the United States (14%, 2017 data), the United Kingdom (13%), Poland (16%), and France (17%) all exhibit relatively high dropout rates [2]. These nations enjoy a high international reputation for their quality of higher education; however, due to strict academic elimination mechanisms, the pressure regarding student dropout remains significant.

In comparison, while the quality of higher education in China has not yet reached the level of the aforementioned countries, scholarly statistics indicate that the dropout rate has long been maintained at a low level. This anomalous association between "low dropout rates" and "low recognition of teaching quality" precisely reflects the deep-seated contradictions within China's education evaluation system: the strict entrance system fails to translate into high-quality educational output during the teaching process.

As scholars such as Wu Daguang have pointed out, "High graduation rates and high employment rates in specific periods by no means represent true high quality; instead, they merely conceal deep-seated quality crises, failing to improve higher education quality and potentially causing damage to it" [3]. Therefore, to reveal the reasons behind the anomalous phenomenon of "low dropout rates" and "low recognition of teaching quality," it is necessary to conduct relevant research, deeply mine and analyze data, and identify the key factors affecting dropout rates.

Against this specific backdrop in China, this paper posits that machine learning technologies offer a methodological solution to the dilemma surrounding educational quality assessment. Although national statistics indicate that the dropout rate remains at a low level, an analysis of the phenomenon suggests that this indicator's anomaly may deviate from the true state of quality due to institutional factors, such as the flexibility of student status management and lenient make-up examination mechanisms.

Constructing a dropout prediction model via machine learning not only enables the micro-level identification of "hidden dropout" groups—defined as students who have not discontinued their studies but remain in a long-term state of academic crisis—but also allows for the reverse deduction of quality shortcomings within the educational process based on risk features. This forms a methodological resonance with the "quality crisis warning" proposed by Professor Wu Daguang.

This study utilizes the "Students Dropout and Academic Success" dataset from the Kaggle platform (containing approximately 4,000 samples). This selection is primarily based on the following three aspects:

First, educational data in China generally suffers from limitations such as unclear recording of dropout situations and short durations of student tracking. Domestic universities typically only publish the total number of dropouts annually without collecting detailed data at the level of individual student behavior, which makes it difficult to establish the feature matrices required for prediction models during experimentation.

Second, for the relatively small-scale dataset used in this paper, traditional machine learning methods possess advantages over deep learning. Machine learning approaches can not only better utilize relational information within limited data through feature engineering but also avoid the overfitting problems that complex neural networks are prone to when data is insufficient.

Third, the learning process data recorded in the educational databases of OECD countries (such as classroom participation, midterm and final grades, etc.) are comparable to data in the academic affairs systems of Chinese universities. This enables the development of a prediction model capable of transferability across different environments. Drawing on the experience of international data allows us to overcome the deficiencies in domestic data collection to a certain extent, providing a feasible technical pathway for future research.

II. Related work

Machine learning methods have now become one of the vital solutions to the dropout challenge. International scholars have conducted extensive research on student prediction algorithms.

A. Current Status of International Research

Adhatrao et al. (2013) [4] proposed a student academic performance prediction system based on ID3 and C4.5 classification algorithms. By integrating student historical data—including entrance exam scores, high school subject scores, gender, and admission type—they constructed a decision tree model used to predict the first-semester pass rate of engineering freshmen. The researchers employed the RapidMiner tool to generate decision trees and combined it with a PHP framework to develop a web application system capable of both single and batch assessments, achieving significant results.

Elakia et al. (2014) [5] further expanded the scope of Educational Data Mining (EDM) by combining student behavior prediction with academic guidance. Like Adhatrao, they utilized ID3 and C4.5 decision tree models. However, unlike the former, Elakia not only recommended post-high school career paths based on student interests and skills but also identified potential violent behavioral tendencies through parameters such as psychological characteristics and extracurricular participation. The study ultimately showed that this model outperformed C4.5 and CHAID algorithms in performance. The authors also highlighted the dual value of EDM techniques in academic planning and early behavioral intervention, suggesting the future introduction of Support Vector Machines (SVM) or neural network technologies to refine prediction granularity.

The findings of Chen et al. (2022) [6] corroborated some previous studies, further confirming the importance of relevant factors and methods in student grade prediction. They found that student academic records and demographic characteristics are significant for grade prediction, while course attributes also have some influence, with other factors having minimal impact. They also investigated the effectiveness of various classification methods. The research indicated that decision tree classifiers are most frequently applied in student grade prediction. Naive Bayes and Random Forest models also performed excellently. Moreover, Random Forest significantly improved prediction performance compared to Naive Bayes and decision trees; additionally, Support Vector Machines (SVM) demonstrated good generalization ability and faster training speeds in small-sample scenarios.

In summary, international research on machine learning related to dropout rates has evolved through multiple stages: progressing from early academic performance predictions based on decision trees (ID3/C4.5), to dropout risk stratification models integrating multi-source data such as psychological and behavioral factors, and eventually to hybrid models utilizing neural networks and support vector machines.

During this process, the research focus has gradually shifted from achieving "risk identification" to "precision intervention." The combination of traditional dropout research models and modern machine learning methods provides reliable theoretical and methodological support for our dropout prediction framework.

B. BGP-Related Routing Data

There is a relative paucity of research by domestic scholars on dropout prediction across the full educational spectrum. However, with the recent rise of online education platforms such as MOOCs, which provide abundant and high-quality training samples, a significant amount of research has been conducted on dropout rates in these online environments. The trend in dropout prediction research has shifted from early phenomenological descriptions to data-driven predictive analysis, and from traditional educational contexts to dropout prediction on new educational platforms. These studies have achieved a degree of success by integrating educational theory with modern machine learning techniques.

In the realm of traditional education, the dropout phenomenon among secondary vocational students is relatively severe. Gao (2010) [7] pointed out that the dropout rate of secondary vocational students is showing an upward trend, with first-year students accounting for approximately 75% of the total number of dropouts. This phenomenon is primarily associated with factors such as financial difficulties, weak academic foundations, and learning weariness. The study further revealed the multidimensional complexity of dropout behavior, involving variables such as gender, major type, and economic background, indicating that dropout rate prediction requires a comprehensive consideration of various factors.

Dropout issues in the field of special education abroad also provide useful references. Wan et al. (2023) [8] analyzed the current status of dropouts among special education students in U.S. high schools. They found that the dropout rate for students receiving special education is not only significantly higher than that of general students but also varies significantly among individuals due to distinct traits such as learning disabilities and emotional disorders. Furthermore, parental expectations and school support have a significant impact on student dropout behavior. This finding provides potential key variables for prediction models, such as student emotional engagement, the intensity of school support, and family educational expectations.

In addition, domestic research has also employed machine learning for dropout rate prediction. Yang et al. (2019) [9] systematically reviewed traditional classification prediction algorithms such as Decision Trees, Support Vector Machines

(SVM), and Neural Networks, laying a foundation for educational data analysis and prediction.

Based on the aforementioned research background, this study aims to implement an optimized K-Means clustering algorithm to group students based on dropout risk and academic performance. By analyzing the centroid characteristics of each cluster without relying on external labels, we aim to explicitly define low, medium, and high-risk student groups. Consequently, personalized intervention suggestions will be formulated, serving as the basis for the development of a proactive Early Warning System (EWS).

III. Data Preprocessing

The dataset utilized in this study is derived from a multi-source database of a university. It encompasses student samples across a wide range of undergraduate disciplines, including agronomy, design, education, nursing, journalism, management, social work, and engineering technologies.

The data captures a diverse array of features recorded at the time of enrollment, comprising academic background, demographic information, and socio-economic factors. Additionally, it documents academic performance metrics at the conclusion of both the first and second semesters. Consequently, this dataset facilitates the construction of a binary classification model for dropout prediction, designed to forecast the likelihood of student dropout versus academic success.

A. Dataset Description

Data collection and screening relevant to the objective of clustering student risk were conducted under the premise of ensuring data correctness. The dataset utilized comprises 4,424 observations of student academic records. Each row represents an individual student profile, rendering the dataset structured and ready for processing within the clustering system.

To rigorously identify the most predictive attributes, feature selection was performed based on Cohen's d effect size. Cohen's d measures the standardized difference between two means, quantifying the separation capability of each feature regarding the target variable. The formula is defined as:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}} \tag{1}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the means of the two groups (e.g., dropout vs. graduate), and s_{pooled} is the pooled standard deviation.

As illustrated in Fig. 1, we calculated the effect sizes for all candidate features. Based on this analysis, the six features with the most significant Cohen's d values were selected as the core attributes for the clustering model, as detailed in Table I.

B. Preprocessing and Transformation

Following initial preprocessing, the data underwent cleaning to remove outliers, thereby enhancing the overall quality of the dataset. Once data cleanliness was verified, transformations were applied, with the most critical being Min-Max normalization.

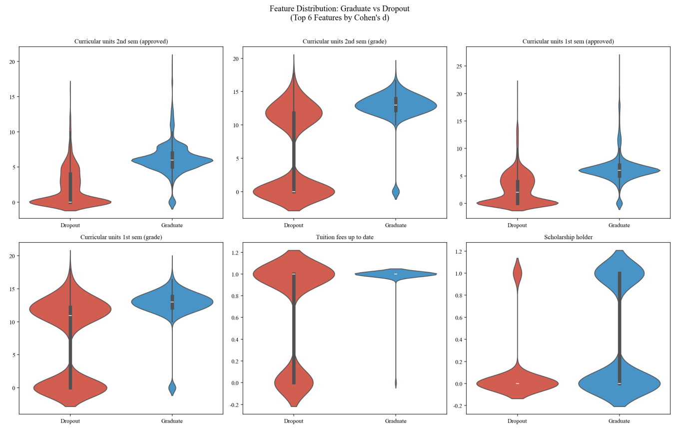

Fig. 1. Cohen's d values indicating the effect size of different features on student dropout.

TABLE I
FEATURES SELECTED IN THIS PAPER

Feature	Mean	Min	Max	Std. Dev.
Curricular units 1st sem (approved)	4.707	0	26.000	3.094
Curricular units 1st sem (grade)	10.641	0	18.875	4.844
Curricular units 2nd sem (approved)	4.435	0	20.000	3.015
Curricular units 2nd sem (grade)	10.230	0	18.851	5.211
Tuition fees up to date	0.881	0	1.000	0.324
Scholarship holder	0.248	0	1.000	0.432

Min-Max normalization was applied to all numerical attributes to scale their values within the range of $[0, 1]$. This step is crucial because the K-Means algorithm is highly sensitive to attributes with differing scales or value ranges. Normalization ensures that each attribute contributes equally to the Euclidean distance calculation, thus preventing attributes with larger numerical magnitudes from dominating the clustering results [10].

The formula for Min-Max normalization is expressed as:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{2}$$

where x is the original value, x' is the normalized value, and $\min(x)$ and $\max(x)$ are the minimum and maximum values of the feature, respectively.

C. Implementation of K-Means Clustering

In the context of data mining, the K-Means clustering algorithm was applied to the normalized dataset. The algorithm operates by partitioning the dataset into K distinct clusters, assigning each data point to the nearest centroid [11]. To ensure algorithmic stability and solution quality, the maximum number of iterations was set to 10.

Experiments were conducted by varying the value of K from 2 to 10. In particular, the configuration yielding the optimal performance was retained for interpretative results. Partitioning the students into three distinct groups (Low Risk, Medium Risk, and High Risk) provides the most applicable argumentative framework for academic management interventions.

IV. Results and Discussion

This section presents the display, analysis, and interpretation of the data. The interpretation encompasses the decoding of data and comparisons with theoretical frameworks and/or similar studies.

A. Optimal Results for Optimizing the Number of Clusters (K)

Determining the optimal number of clusters is a critical configuration in the clustering process, as it directly impacts the structure and quality of the model. The optimization process involved testing a range of values from $K = 2$ to $K = 10$, with each value undergoing 10 iterations to ensure stability.

Cluster quality assessment employed Classification Error and the Davies-Bouldin Index (DBI), where lower values indicate superior clustering quality. The summarized test results for each distinct value are presented in Fig. 2.

As illustrated in Fig. 2, when $K = 9$, the Classification Error is minimized and the DBI value is relatively low, indicating optimal clustering quality. Therefore, $K = 9$ was established as the final value for the clustering model.

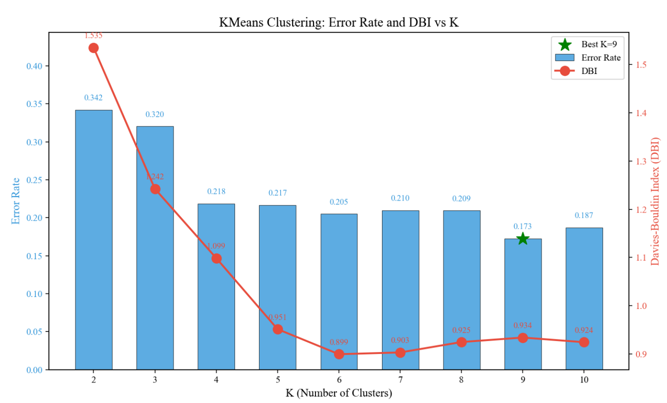

Fig. 2. Performance metrics (Classification Error and DBI) for different K values.

B. Visualization of Clustering Results

To validate the quality of the clustering and visualize the structure of the high-dimensional data, we employed t-Distributed Stochastic Neighbor Embedding (t-SNE) [12] to project the six-dimensional feature space into a three-dimensional perceptible space.

As shown in Fig. 3, the 4,424 student samples are plotted in a 3D space, where the axes represent the three primary components derived from the t-SNE algorithm. The distinct colors correspond to the $K = 9$ clusters identified by the K-Means algorithm.

The visualization results indicate the following:

- **Clear Separation:** The nine clusters exhibit significant spatial separation in the projected space. There are clear boundaries between most groups, suggesting that the K-Means algorithm successfully captured the intrinsic distinctions within the student data structure.

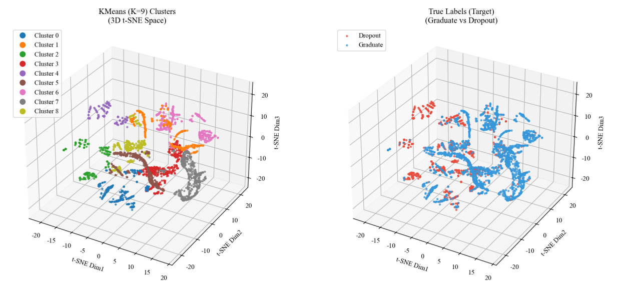

Fig. 3. 3D visualization of student clusters using t-SNE dimensionality reduction.

- **Cluster Compactness:** Points within the same color group tend to aggregate closely. This high intra-cluster compactness combined with inter-cluster separation confirms that the choice of $K = 9$ provides a robust partitioning of the dataset, rather than arbitrarily cutting through a continuous distribution.
- **Validation of K-Means:** The t-SNE projection visually corroborates the quantitative metrics (DBI and Classification Error) discussed earlier. Even after reducing the dimensions, the heterogeneity of the student profiles remains preserved and distinguishable, justifying the use of a fine-grained clustering approach ($K = 9$) over a simpler coarse-grained model.

C. Prediction of Dropout Rates and Intervention for Enrolled Students

Based on the clustering results ($K = 9$), we analyzed the composition of each cluster regarding the target variable (Dropout, Enrolled, Graduate). This step is crucial for translating unsupervised patterns into actionable intervention strategies. As illustrated in Fig. 4 and Fig. 5, the clusters exhibit distinct risk profiles, which we categorize into High Risk and Medium Risk groups to guide specific interventions.

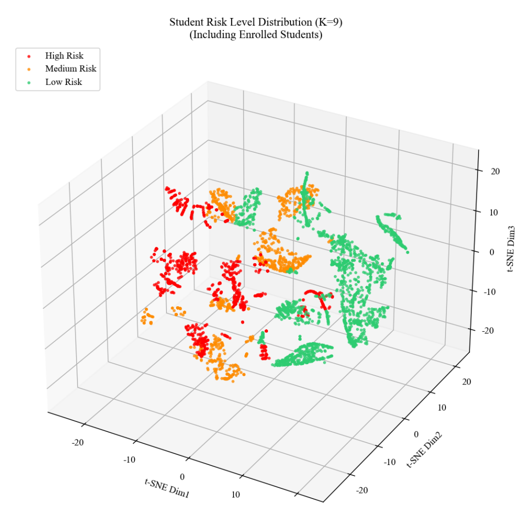

Fig. 4. Distribution of target categories (Dropout, Enrolled, Graduate) across the nine clusters.

Cluster Risk Analysis (K=9)
Dropout Rate = Dropout / (Dropout + Graduate)

Cluster	Total	Dropout	Graduate	Enrolled	Dropout Rate	Risk Level
3	401	380	3	18	0.9922	High Risk
2	624	279	100	245	0.7361	High Risk
5	506	326	130	50	0.7149	High Risk
1	591	121	306	164	0.2834	Medium Risk
4	373	75	208	90	0.265	Medium Risk
8	665	92	496	77	0.1565	Low Risk
6	329	38	206	85	0.1557	Low Risk
0	422	49	336	37	0.1273	Low Risk
7	513	61	424	28	0.1258	Low Risk

Fig. 5. Detailed risk assessment for Enrolled students within specific clusters.

1) High-Risk Group (Red Zone): The clusters dominated by the "Dropout" category (represented in red in the figures) are identified as High-Risk groups. Students falling into these centroids exhibit the most critical warning signs, such as extremely low grade point averages (GPA) combined with unpaid tuition fees.

For this group, the prediction indicates a near-certainty of dropout without immediate external interference.

- **Intervention Strategy:** Implement an "Emergency Intervention Plan." This includes immediate financial counseling for tuition debt restructuring and mandatory academic tutoring. The university administration should prioritize these students for face-to-face interviews to identify non-academic barriers (e.g., mental health or family issues).

2) Medium-Risk Group (Yellow Zone): The clusters characterized by a significant proportion of "Enrolled" status (represented in yellow) constitute the Medium-Risk group. Unlike the high-risk group, these students have not yet dropped out but are failing to graduate on time. They represent the "hidden dropout" phenomenon—students who are technically enrolled but academically stagnant.

The analysis suggests these students are often passing some courses but failing critical ones, or lacking the motivation to complete final requirements.

- **Intervention Strategy:** Deploy a "Preventive Support System." Interventions should focus on academic planning, such as course load adjustment and mentorship programs. The goal is to transition them from a state of stagnation (Enrolled) to completion (Graduate), preventing them from sliding into the High-Risk category in subsequent semesters.

By distinguishing between the "Red" (acute crisis) and "Yellow" (chronic stagnation) signals, the proposed framework allows the university to allocate resources more efficiently, moving from a passive recording of dropout statistics to proactive quality assurance.

V. Conclusion

This paper addresses the anomaly between "low dropout rates" and "low teaching quality recognition" in the context of Chinese higher education. By shifting the perspective from traditional supervised classification to unsupervised clustering, we successfully constructed a data-driven framework for identifying student dropout risks.

The study demonstrates that a simple binary distinction (Dropout vs. Graduate) is insufficient for capturing the complexity of student academic trajectories. Through the implementation of K-Means clustering with an optimized $K = 9$, we uncovered distinct subgroups, particularly identifying the "Medium-Risk" enrolled students who are often overlooked in traditional statistics. The visualization via t-SNE and the feature selection based on Cohen's d further validated the robustness of the model.

In conclusion, the "quality crisis" pointed out by scholars can be mitigated by leveraging machine learning to reveal hidden patterns. The identified "Red" and "Yellow" risk zones provide a concrete roadmap for university administrators to implement personalized interventions. Future work will focus on integrating this model into a real-time academic affairs system to enable dynamic monitoring and timely support, ultimately bridging the gap between enrollment expansion and educational quality.

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